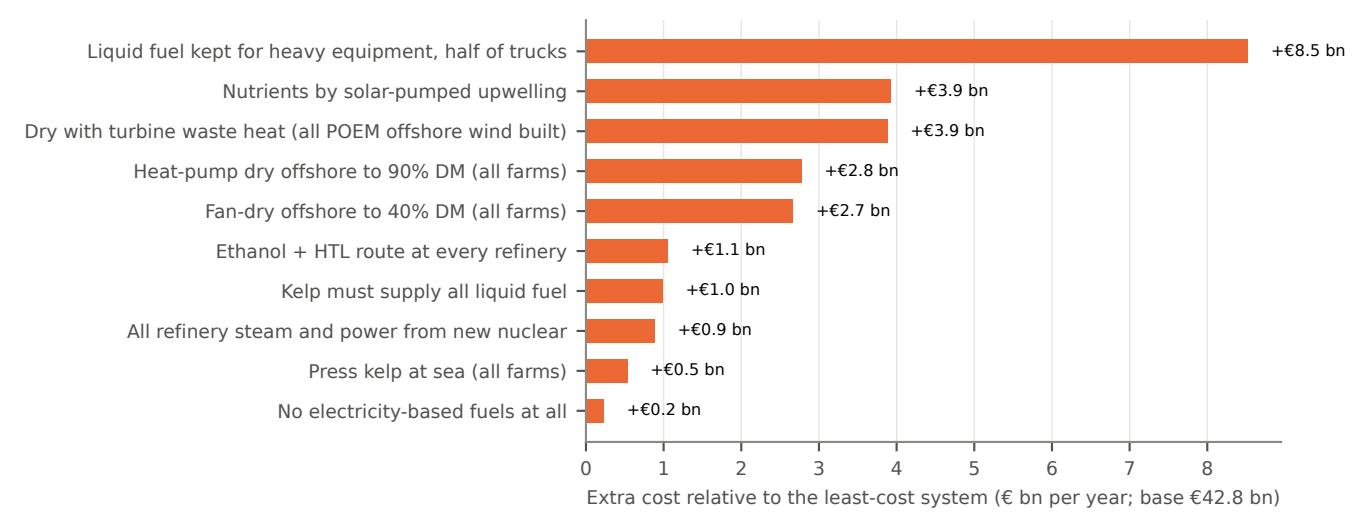
Liquid fuel kept for heavy equipment, half of trucks
+€8.5 bn
Nutrients by solar-pumped upwelling
+€3.9 bn
Dry with turbine waste heat (all POEM offshore wind built)
+€3.9 bn
Heat-pump dry offshore to 90% DM (all farms)
+€2.8 bn
Fan-dry offshore to 40% DM (all farms)
+€2.7 bn
Ethanol + HTL route at every refinery
+€1.1 bn
Kelp must supply all liquid fuel
+€1.0 bn
All refinery steam and power from new nuclear
+€0.9 bn
Press kelp at sea (all farms)
+€0.5 bn
No electricity-based fuels at all
+€0.2 bn
0
1
2
3
4
5
6
7
8
Extra cost relative to the least-cost system (€ bn per year; base €42.8 bn)